

Template for Computer Programming I Practical Recording

Date	:	20/02/2025
Problem Specification	:	The problem trying to find square root of given number.
Assumption	:	User input is valid.
Limitation	:	input $a \geq 0$ Random value $x_0 \neq 0$
Input	:	user input the number n
Processing	:	calculate the square root of given number n using given format
Output	:	output square root of given number n in 6 digit decimal number
Algorithm	:	Step1: Start Step2: Prompt as "Enter the number a" Step3: Read the value for a from the user Step4: if ($a \geq 0$) is false proceed to step 15 Step5: Declare variable $x_0 \leftarrow 0$ Step6: Repeat Step 7, Step 8 until ($x_0 = 0$) Step7: Get the random value for x_0 Step8: Declare $x_0 \leftarrow x_0$ Step9: Declare $x \leftarrow x_0$ Step10: Repeat Step 11, Step 12 until ($x * x - a > 0.0000001$) Step11: declare variable x_1 : $x_1 \leftarrow (x + a/x)/2$ Step12: $x \leftarrow x_1$ Step13: if ($x > 0$) is true Print x as a six-digit decimal value, then go to Step16 Step14: else Print ($x * (-1)$) as a six-digit decimal value then go to Step 16 Step15: Print as "wrong input, Input a positive number or 0" Step16: End the Programme.
Programme listing	:	Programme file attached
Test data and expected output	:	1. Test data : a=15 Expected output : 3.872983
Output obtained for test data	:	1. Test data : a=15 Obtained output : 3.872983
Discussion	:	Take an idea on how to use while loops and set the range correctly.

Conclusion

: The process ensures an accurate estimate of the square root. Finally, the result is displayed as a six-decimal value, providing a precise output.